

TEAM ASTERIX

Electronics & Powertrain

WORKSHOP 2026

Details

Duration	7 October 2026 - 2 November 2026 (4 weeks, 12 core sessions), followed by 2 complimentary sessions on 4 and 6 November 2026
Schedule	Monday, Wednesday and Friday
Timing	5:15 PM - 6:45 PM (1.5 hours per session) for all sessions
Audience	First-year students. No prior knowledge needed, we start from scratch!
Subjects	Network Analysis, Electronic Devices, Microcontrollers, Analog Circuits, Electric Machines & Motors, PCB Design
Format	9 talks, 3 hands-on sessions on the computer and 2 bonus sessions (Mechanical and Software)

Workshop Objectives

- Learn the basics of circuits, electronic parts, microcontrollers and motors, giving you a head start for your second semester and second year.
- See how the electronics inside the Team ASTERIX autonomous buggy actually work.
- Try things out yourself using free circuit simulators and a PCB design tool.
- Get a peek at the mechanical and software side of the buggy in the bonus sessions.

The syllabus [High level overview]

- Network Analysis
- Electronic Devices
- Microcontrollers
- Analog Circuits
- Electric Machines & Motors
- PCB Design

Workshop Structure

The workshop combines lectures, guided computer-based hands-on sessions, and two complimentary sessions that connect electronics with the mechanical and software sides of the autonomous buggy.

Timetable

Session Timing: 5:15 PM – 6:45 PM

Week	Monday	Wednesday	Friday
Week 1 · 5 - 9 Oct	No session	7 Oct 2026 Network Analysis Circuit Basics: How Electricity Flows Niyathi & Akshayaa	9 Oct 2026 Electronic Devices Diodes & Transistors: The Tiny Switches Niyathi & Kathin
Week 2 · 12 - 16 Oct	12 Oct 2026 Hands-on · Electronic Devices Hands-on 1: Simulating a Transistor Switch Handled jointly by the team	14 Oct 2026 Microcontrollers Meet the Microcontroller: ESP32 Basics Vishal & Aravind	16 Oct 2026 Analog Circuits How Our Buggy Starts Up Allwin (Alumnus) & Akshayaa
Week 3 · 19 - 23 Oct	19 Oct 2026 Hands-on · Analog Circuits / Microcontrollers Hands-on 2: Build the Buggy's Start-Up in Tinkercad Handled jointly by the team	21 Oct 2026 Analog Circuits Changing Voltages: Buck & Boost Converters Priyanka & Niyathi	23 Oct 2026 Analog Circuits Op-Amps & Filters Made Simple Akshayaa & Kathin
Week 4 · 26 - 30 Oct	26 Oct 2026 Electric Machines & Motors How Electric Motors Work Aravind & Vishal	28 Oct 2026 PCB Design PCB Design I: What Is a PCB and How Is It Made? Saswin (Alumnus) & Nanthavikraman (Alumnus)	30 Oct 2026 PCB Design PCB Design II: Laying Out a Board Saswin (Alumnus) & Nanthavikraman (Alumnus)
Week 5 · 2 - 6 Nov	2 Nov 2026 Hands-on · PCB Design Hands-on 3: Design Your First PCB Saswin (Alumnus) & Nanthavikraman (Alumnus)	4 Nov 2026 Complimentary Session Mechanical Basics Team ASTERIX	6 Nov 2026 Complimentary Session Software Basics + Valedictory Team ASTERIX

Subject Key

Network Analysis · Electronic Devices · Microcontrollers · Analog Circuits · Electric Machines & Motors · PCB Design · Hands-on / Simulation · Complimentary Session

SESSION 1 · NETWORK ANALYSIS

Circuit Basics: How Electricity Flows

Date	Wednesday, 7 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Niyathi & Akshayaa	Type	Lecture

Topics covered

- The basics: voltage, current, resistance and power, and how they relate
- Ohm's law and combining resistors in series and parallel
- Kirchhoff's laws (KCL & KVL) to solve simple circuits
- How capacitors charge and discharge in RC circuits, and where we use them

By the end, you'll be able to:

- Use Ohm's law and Kirchhoff's laws on simple circuits
- Explain how a capacitor charges and discharges

SESSION 2 · ELECTRONIC DEVICES

Diodes & Transistors: The Tiny Switches

Date	Friday, 9 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Niyathi & Kathin	Type	Lecture

Topics covered

- What semiconductors are and how a diode lets current flow only one way
- How transistors (BJT and MOSFET) work
- Using a MOSFET as an electronic switch
- Why MOSFETs are the go-to choice in power circuits

By the end, you'll be able to:

- Explain what a diode does
- Use a MOSFET as a simple on/off switch

SESSION 3 · HANDS-ON · ELECTRONIC DEVICES

Hands-on 1: Simulating a Transistor Switch

Date	Monday, 12 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Handled jointly by the team	Type	Hands-on (Simulation)

Topics covered

- Getting started with PSpice / LTspice: placing parts and wiring a circuit
- Building a simple MOSFET test circuit
- Running simulations and plotting how the MOSFET behaves
- Bonus: simulating a MOSFET switching a load on and off

By the end, you'll be able to:

- Build and run a basic circuit simulation
- Read a graph that shows how a MOSFET behaves

SESSION 4 · MICROCONTROLLERS

Meet the Microcontroller: ESP32 Basics

Date	Wednesday, 14 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Vishal & Aravind	Type	Lecture + Demo

Topics covered

- What a microcontroller is and why modern vehicles are full of them
- Meet the ESP32: its main features and pins
- Setting up Arduino IDE and writing your first program
- First projects: blinking an LED, reading a button and dimming an LED

By the end, you'll be able to:

- Name the main parts of a microcontroller like the ESP32
- Upload your first program and make an LED blink

Before the session: Install Arduino IDE. Create a free Tinkercad account for the hands-on session on 19 Oct.

SESSION 5 · ANALOG CIRCUITS

How Our Buggy Starts Up

Date	Friday, 16 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Allwin (Alumnus) & Akshayaa	Type	Lecture

Topics covered

- A tour of the buggy's electrical system, from battery to loads
- The starting sequence: how the buggy powers up, step by step
- Safety and kill switches that keep everything under control
- How small control signals switch big loads, plus lessons from competition

By the end, you'll be able to:

- Walk through the buggy's start-up steps
- Explain how a small signal can switch on a big load

SESSION 6 · HANDS-ON · ANALOG CIRCUITS / MICROCONTROLLERS

Hands-on 2: Build the Buggy's Start-Up in Tinkercad

Date	Monday, 19 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Handled jointly by the team	Type	Hands-on (Simulation)

Topics covered

- Getting to know Tinkercad Circuits and the Arduino Uno
- Building the buggy's starting sequence with buttons, LEDs and a motor
- Coding the sequence, including safety checks and an emergency stop
- Testing and debugging everything in simulation

By the end, you'll be able to:

- Code a simple start-up sequence with safety checks
- Test your circuit and code in simulation

SESSION 7 · ANALOG CIRCUITS

Changing Voltages: Buck & Boost Converters

Date	Wednesday, 21 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Priyanka & Niyathi	Type	Lecture

Topics covered

- Why a vehicle needs different voltage levels for its electronics
- Linear vs switching regulators: which one wastes less energy
- How buck (step-down) and boost (step-up) converters work
- Ready-made converter modules and where they're used in the buggy

By the end, you'll be able to:

- Explain how buck and boost converters step voltage down or up
- Tell when to use a regulator and when to use a converter

SESSION 8 · ANALOG CIRCUITS

Op-Amps & Filters Made Simple

Date	Friday, 23 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Akshayaa & Kathin	Type	Lecture

Topics covered

- What an op-amp is and the simple rules it follows
- Common op-amp circuits: amplifiers, buffers and comparators
- Why we need filters to clean up noisy sensor signals
- Building simple low-pass and high-pass filters

By the end, you'll be able to:

- Recognise basic op-amp circuits
- Explain how a simple filter removes unwanted noise

SESSION 9 · ELECTRIC MACHINES & MOTORS

How Electric Motors Work

Date	Monday, 26 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Aravind & Vishal	Type	Lecture

Topics covered

- How electricity and magnetism work together to make motors spin
- DC motors and how we control their speed and direction
- BLDC, stepper and servo motors, and where each one is used
- The motor and controller used in our buggy

By the end, you'll be able to:

- Explain how a motor turns electricity into motion
- Tell apart DC, BLDC, stepper and servo motors

SESSION 10 · PCB DESIGN

PCB Design I: What Is a PCB and How Is It Made?

Date	Wednesday, 28 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Saswin (Alumnus) & Nanthavikraman Type (Alumnus)		Lecture + Demo

Topics covered

- What a PCB is and why we use it instead of a breadboard
- Parts of a PCB: tracks, pads, the green coating and the white labels
- Through-hole vs surface-mount components
- The design journey from circuit diagram to finished board, with a software demo

By the end, you'll be able to:

- Identify the main parts of a PCB
- Describe, in simple steps, how a circuit diagram becomes a finished board

SESSION 11 · PCB DESIGN

PCB Design II: Laying Out a Board

Date	Friday, 30 Oct 2026	Time	5:15 PM – 6:45 PM
Handled by	Saswin (Alumnus) & Nanthavikraman Type (Alumnus)	Lecture + Demo	

Topics covered

- Turning a circuit diagram into a board layout
- Placing parts neatly and drawing tracks the right way
- Checking the design for mistakes before the board is made
- A tour of the PCB inside the Team ASTERIX buggy

By the end, you'll be able to:

- Follow the basic rules for placing parts and drawing tracks
- Point out the main sections of the buggy's PCB

SESSION 12 · HANDS-ON · PCB DESIGN

Hands-on 3: Design Your First PCB

Date	Monday, 2 Nov 2026	Time	5:15 PM – 6:45 PM
Handled by	Saswin (Alumnus) & Nanthavikraman Type (Alumnus)	Hands-on (Guided Demonstration)	

Topics covered

- Follow along as the instructors design a simple LED board
- Drawing the circuit and placing the parts
- Connecting the tracks, running the design check and fixing mistakes
- Seeing the finished board in 3D, plus Q&A;

By the end, you'll be able to:

- Follow the complete design process for a very simple circuit

SESSION C1 · COMPLIMENTARY SESSION

Complimentary Session: Mechanical Basics

Date	Wednesday, 4 Nov 2026	Time	5:15 PM – 6:45 PM
Handled by	Team ASTERIX	Type	Complimentary Session

Topics covered

- A look at the buggy's mechanical parts: frame, suspension, steering, brakes and gears
- How the electronics and the mechanical parts work together

By the end, you'll be able to:

- See how the mechanical and electrical parts of the buggy fit together

SESSION C2 · COMPLIMENTARY SESSION

Complimentary Session: Software Basics + Valedictory

Date	Friday, 6 Nov 2026	Time	5:15 PM – 6:45 PM
Handled by	Team ASTERIX	Type	Complimentary Session + Valedictory

Topics covered

- A look at the software that drives the autonomous buggy, plus programming basics
- Valedictory: workshop wrap-up, feedback and closing

By the end, you'll be able to:

- Understand what software does in the autonomous buggy

Subject-wise Summary

Subject	Sessions	Dates	Hours
Network Analysis	1	7 Oct	1.5 h
Electronic Devices	1	9 Oct	1.5 h
Microcontrollers	1	14 Oct	1.5 h
Analog Circuits	3	16, 21, 23 Oct	4.5 h
Electric Machines & Motors	1	26 Oct	1.5 h
PCB Design	2	28, 30 Oct	3.0 h
Hands-on / Simulation	3	12 Oct (PSpice/LTspice), 19 Oct (Tinkercad), 2 Nov (PCB)	4.5 h
Complimentary Session	2	4 Nov (Mechanical), 6 Nov (Software + Valedictory)	3.0 h

Total: 14 sessions · 12 core sessions + 2 complimentary sessions · 21.0 hours

Student Timetable

Electronics & Powertrain Workshop 2026 · Team ASTERIX · aBAJA SAEINDIA

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Bring a laptop with the required software installed to the hands-on sessions on 12 Oct, 19 Oct and 2 Nov. We'll help you set it up!